

PFSM – Plate-solving push-to, on a full imaging rig

What PFSM adds between the two –
for PiFinder and StellarMate developers

Community project — not affiliated with PiFinder or StellarMate

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Astronomische Vereinigung
Vorderpfalz e.V.

Faszination. Astronomie. Erleben.

Agenda

Introduction

For newcomers

Project summary & focus

Implementation

Control Center

PF LX200 & Mount Bridge

PF Simulator

Contribute

Upstream: PiFinder

App: StellarMate

Where we are

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Two problems, one telescope

Where am I pointing?

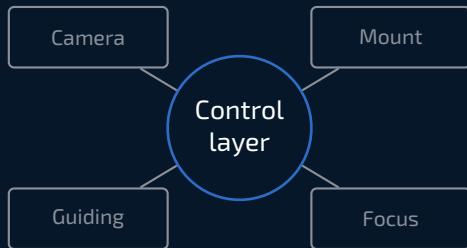
- ◆ A Dobson or push-to EQ has no encoders.
- ◆ The software has no idea where the tube looks.



no position feedback

Drive the whole rig

- ◆ Camera, mount, guiding, focus.
- ◆ All need one control layer: INDI, KStars/Ekos, an app.



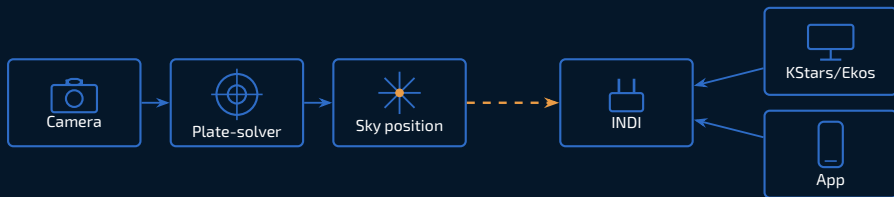
Each project owns one half

PiFinder

- ◆ Global-shutter camera + plate-solver.
- ◆ Live sky position + push-to arrows.
- ◆ Python, open hardware & software.

StellarMate

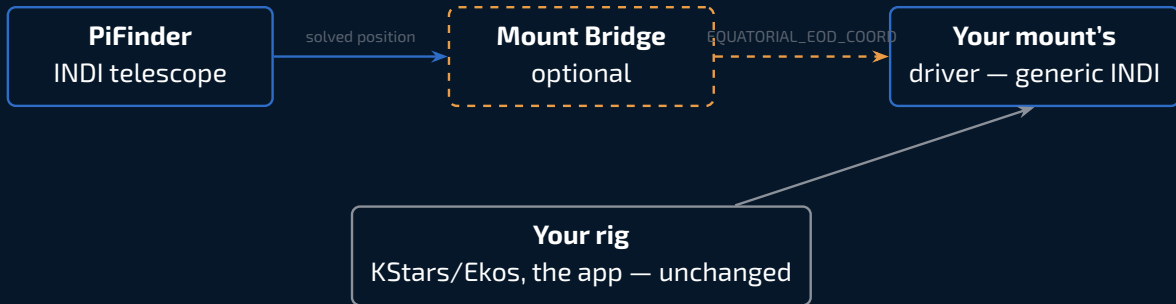
- ◆ INDI — the device-driver standard.
- ◆ KStars / Ekos — capture & automation.
- ◆ The StellarMate phone / tablet app.



PFSM is the wiring. INDI is the connector.

- ◆ **PiFinder** exposes its solved position as an INDI telescope.
- ◆ **Mount Bridge** couples that to any INDI mount (optional).
- ◆ **Your rig** — KStars / Ekos, the mount, the app — stays unchanged.

PiFinder is patched in place, never forked. The mount side is generic INDI.



What PFSM is

- ◆ **A setup script** — installs & patches PiFinder onto StellarMate OS.
- ◆ **diffs/*.diff patches** — applied to a stock checkout via `patch(1)`.
- ◆ **Three INDI drivers** — LX200, Mount Bridge, Simulator.
- ◆ **A web Control Center** — stdlib-only, port 8765.

What PFSM is not

- ◆ **Not a fork of PiFinder** — patches re-applied every update, kept minimal.
- ◆ **Not a fork of StellarMate** — stock INDI, Web Manager, KStars.
- ◆ **Not a replacement for the INDI Control Panel** — it stays for advanced work.
- ◆ **Not PiFinder-version-specific** — talks only to the stable `/api/*`.

Why INDI

Both sides already speak it.

- ◆ PiFinder exposes its position as an INDI telescope.
- ◆ The Mount Bridge couples that to any INDI mount — no per-mount code.

Versions & platforms

PiFinder 2.6.3 · StellarMate OS 2.3.0

- ◆ Pi 4 — fully tested.
- ◆ Pi 5 — fully tested.
- ◆ UTM x86 — install + Control Center + Mount Bridge vs. the Simulator (no real solving).

Pinned to a fixed release tag, not the upstream *release* branch.

Prerequisites

- ◆ **A current StellarMate OS (SMOS) installation** — the only supported base.
- ◆ **Not supported: a stock PiFinder OS image.** Doesn't make sense either — PFSM's entire premise is the StellarMate integration (INDI, KStars/Ekos, the Web Manager, the app), and a stock PiFinder OS has none of that to integrate with.

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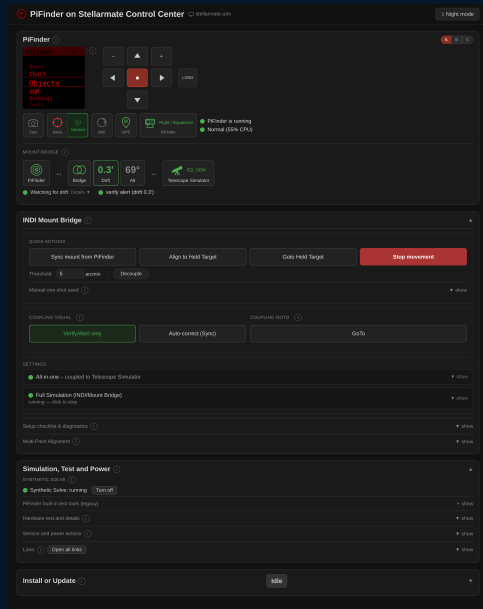
Contribute

Upstream: PiFinder

App: StellarMate

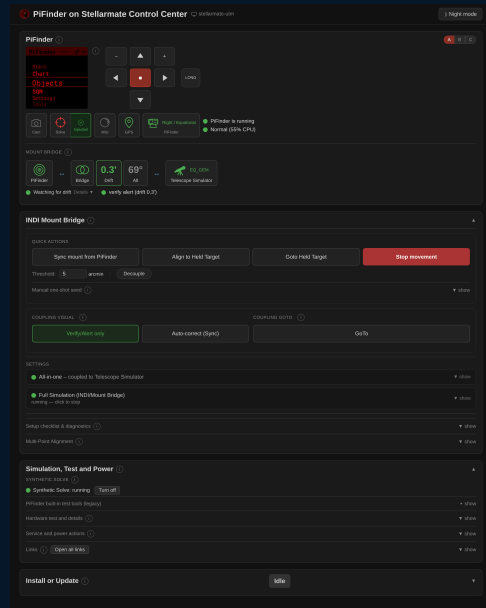
Why a Control Center

- ◆ Watch an install from a phone at the scope.
- ◆ Switch Fake Mode ↔ the real service.
- ◆ Check camera / IMU / GPS are really detected.
- ◆ Reboot the Pi without SSH.

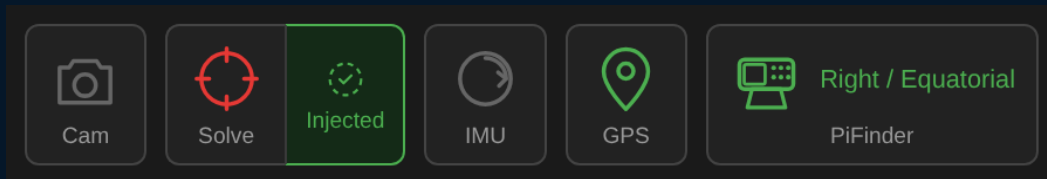


Stdlib only – by necessity

- ✦ It bootstraps the very venv that PiFinder needs, so it can't depend on one.
- ✦ `http.server` on `:8765`, polled every 1–2 s.
- ✦ HTTP Basic auth against the system account (PAM).



Traffic-light badges

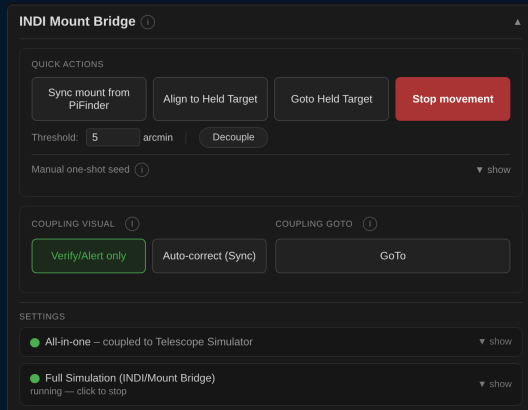


- ◆ **White** — not applicable yet.
- ◆ **Green** — verified functional.
- ◆ **Yellow** — degraded / estimate.
- ◆ **Red** — broken or wrong.

The rule: verify real state, never “the process is alive”.

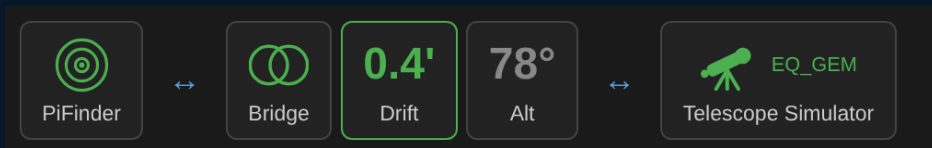
Coupling, without the INDI panel

- ◆ Quick Actions — one-shots.
- ◆ Coupling presets — one click.
- ◆ Settings — set once, hidden.



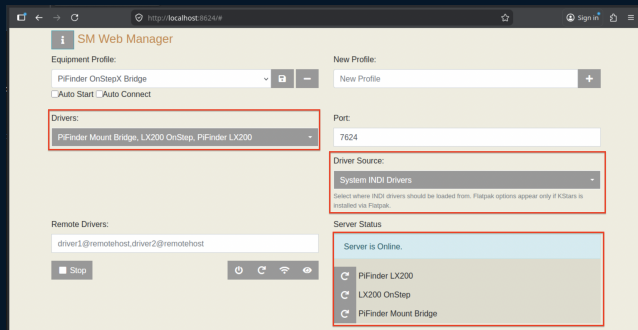
Three building blocks

1. **PiFinder LX200** — INDI telescope. Reports the solved position; a GoTo becomes a push-to. Required.
2. **PiFinder Mount Bridge** — optional. Snoops PiFinder + the mount and couples them. Generic INDI only.
3. **Your mount's driver** — not ours. Only for a motorised mount.



The Web Manager – a “Remote” profile

- ◆ PFSM's INDI drivers live **only** in the StellarMate Web Manager (:8624) — the Equipment Profile is built there.
- ◆ KStars / Ekos then connects in **Remote Host** mode, never “Local”.
- ◆ The one real difference from a stock StellarMate setup.



One weak PiFinder, one strong box

Pick the **role** at setup:

All-in-one

PiFinder hardware with mount.

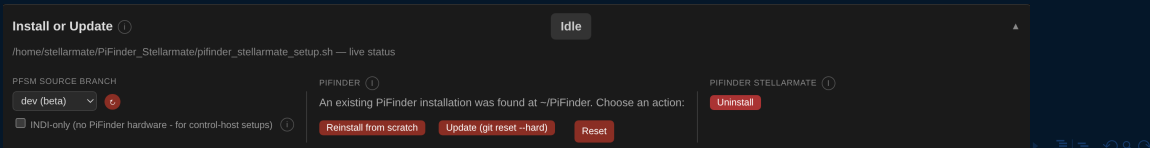
PiFinder host

PiFinder only, no mount.

Control host

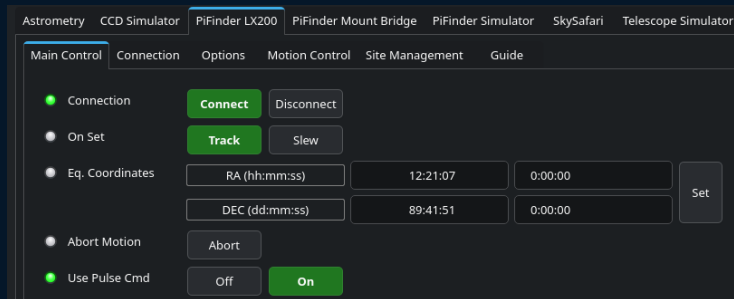
No PiFinder hardware. Couples to remote PiFinder.

- ◆ **Control host** — a strong StellarMate PC runs INDI, Ekos, guiding and capture; the low-power PiFinder Pi over the LAN only supplies its solved position.
- ◆ **INDI-only install** (`--mode=indi_only`) — the Control-host case: just the INDI deps + the two PFSM drivers, no PiFinder clone or venv.



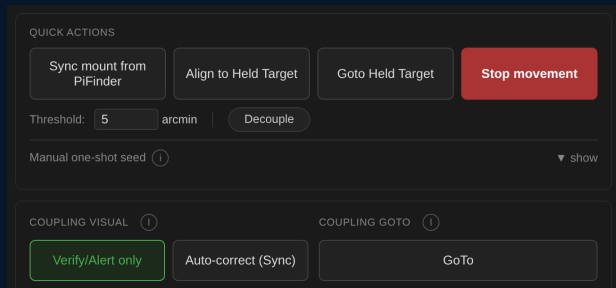
PiFinder LX200

- ✦ Talks LX200 to PiFinder's `pos_server.py` (TCP 4030).
- ✦ Reports the solved position each poll; a GoTo is only a push-to.
- ✦ No Sync — PiFinder measures the truth every frame.



The Coupling Dial

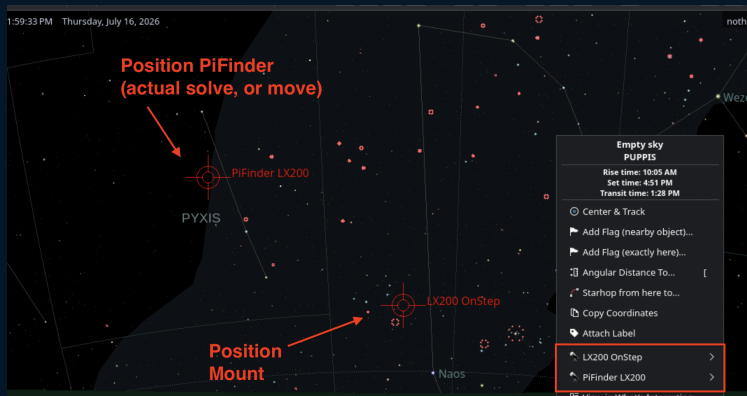
1. **Off**
pure push-to
2. **Verify / Alert only**
warn on drift, hands off
3. **Auto-correct on drift**
Sync past the threshold
4. **Goto-Forward**
real Goto, then settle & refine



Generic INDI, nothing mount-specific

Only EQUATORIAL_EOD_COORD + ON_COORD_SET — never a mount command.

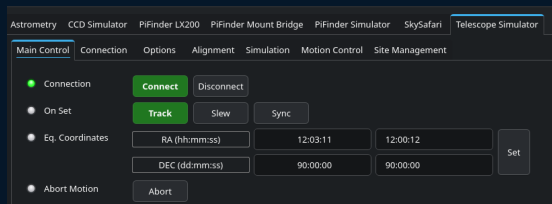
- ◆ Multi-Point Alignment (#191) — automated multi-star run.
- ◆ Reposition detection — adopt or revert a move.
- ◆ Shadow Sync — mirror onto a non-driving device.



A mount side and a PiFinder side

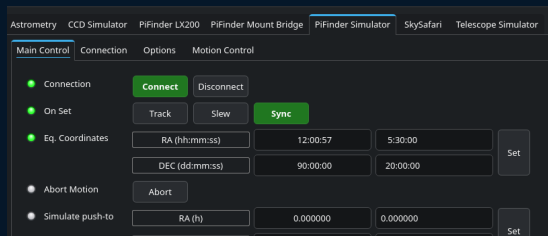
Mount side

Stock INDI Telescope Simulator —
Sync or GoTo like a real mount.



PiFinder side

- ◆ **Simulator** — holds a settable RA/Dec.
- ◆ **Truth Injector** — POSTs to /api/fake_solve.



Simulator details

- ◆ **Sync == Goto** — no motion to model; the RA/Dec just becomes the held value.
- ◆ **Inject every cycle** — keeps PiFinder's solve timestamp fresh.
- ◆ **FOLLOW_MOUNT_DEVICE (#239)** — optionally dead-reckon along a mount's slews.

Next (v3.x): render a GSC star field and run PiFinder's real solver on it — issue #344.

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PiFinder – patches that aren't PFSM-specific

docs/upstream_patch_inventory.md

- ◆ **Multi-client** `pos_server.py` — a thread per connection. It served one client at a time; SkySafari, Stellarium, KStars and our LX200 driver all want it at once. (docs/concepts/pos_server_multi_client_architecture.md)
- ◆ Test Mode toggle + status readback — a reliable API trigger.
- ◆ Camera-process resilience — fall back to the debug camera instead of crashing.
- ◆ `all_ips()` — show every network address, not just the default route.
- ◆ LX200 “no position yet” — return `None`, not a parseable fake coordinate.
- ◆ Near-pole `/api/fake_solve RA` — filed as brickbots/PiFinder#645.

Ready-to-file PR text: docs/upstream_pr_templates.md.

StellarMate – extending the SMOS app for PFSM

What a thin integration needs:

- ◆ Start / stop the selected INDI profile.
- ◆ Add / remove just the PiFinder drivers.
- ◆ Pick “the mount” from the loaded drivers.
- ◆ Three Coupling presets, in PiFinder’s words.

The coupling logic is already framework-agnostic Python — a thin adapter ports it.

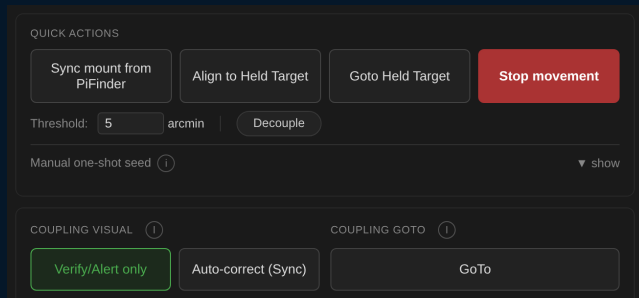
Concept: `docs/concepts/mount_bridge_web_integration.md`.

[[feature → app screen

- ◆ Status badges → a read-only status card.
- ◆ Coupling presets → a segmented control.
- ◆ Quick Actions → buttons on that card.

APIs already exist:

/api/mount_bridge_*, or the
INDI properties directly.



Roadmap & where to jump in

v2.x — consolidate

- ◆ Harden & test what exists.
- ◆ Send selected patches upstream.
- ◆ A guiding watcher.

v3.x — integrate

- ◆ Deeper SMOS integration (#35, #36).
- ◆ PFSM actions in the PiFinder app (#43).
- ◆ A “real” GSC-solving simulator (#344).

github.com/apos/PiFinder_Stellarmate · GitHub Project 15

Two takeaways

1. **PiFinder devs** — a triaged inventory of patches that belong upstream, PR text already written.
2. **StellarMate devs** — the SMOS app could drive basic PFSM use with a thin adapter over existing APIs.

◆ Outlook

- ◆ Real-sky solver simulation (#344).
- ◆ Selected patches back to brickbots/PiFinder.
- ◆ App-side coupling controls.

Thanks – and clear skies.



. PiFinder StellarMate

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Unofficial community project — not affiliated with StellarMate or PiFinder.

Links

- ◆ github.com/apos/PiFinder_Stellarmate — repo, README, docs.
- ◆ [docs/upstream_patch_inventory.md](#) / [docs/upstream_pr_templates.md](#)
- ◆ [docs/concepts/mount_bridge_web_integration.md](#)
- ◆ [brickbots/PiFinder](#) — upstream · [issue #645](#) (near-pole RA)
- ◆ [GitHub Project 15](#) — tracked roadmap.